# Customer Revenue, Risk & Retention Analytics Project

## Project Objective

This project simulates a real-world analytics problem faced by e-commerce companies. Students will analyze customer behavior, revenue patterns, returns, and delivery performance using Python and Pandas. The focus is on logic building, data manipulation, and business insight generation.

## Datasets Provided

1. customers.csv – demographic and signup information
2. orders.csv – order lifecycle and payment details
3. order_items.csv – item-level purchase information

## Mandatory Column Creation (Very Important)

1. order_value – calculated using quantity × price (function mandatory)
2. delivery_delay_days – difference between delivery_date and order_date
3. is_repeat_customer – using lambda
4. is_high_value_order – order_value > 3000 using lambda
5. customer_value_segment – High / Medium / Low using if–else logic
6. customer_risk_category – based on return rate using if–else
7. order_reliability_score – Delivered (+1), Returned (–1), Cancelled (–2)

## Programming Constraints

1. At least 3 custom Python functions must be created
2. Lambda expressions must be used for logical column creation
3. At least one for-loop must be used (groupby-only solutions not allowed for that task)
4. Blind dropping of missing values is not allowed
5. Every visualization must have a business explanation

## Business Questions to Answer

1. Which customers generate high revenue but have high return rates?
2. Which cities suffer most from delivery delays and revenue loss?
3. Are repeat customers actually more valuable?
4. Which categories appear profitable but lose revenue due to returns?
5. Which payment mode carries the highest risk?

## Visualization Requirements

1. Monthly revenue trend – Line chart
2. Customer value segmentation – Bar chart
3. Return rate by category – Bar chart

4. Delivery delay vs revenue – Scatter plot
5. City vs category revenue – Heatmap

## Final Submission Guidelines

Students must submit a Jupyter Notebook containing clean code, derived columns, visualizations, and an insights section. The final summary must include three business problems identified and three data-backed recommendations.